

A Tool for Monitoring and Controlling Standalone Immersive HCI Experiments

Alejandro Rey López

alejandro.rey@uc3m.es

Universidad Carlos III de Madrid
Leganés, Madrid, Spain

Paloma Díaz Pérez

pdp@inf.uc3m.es

Universidad Carlos III de Madrid
Leganés, Madrid, Spain

Andrea Bellucci

abellucci@inf.uc3m.es

Universidad Carlos III de Madrid
Leganés, Madrid, Spain

Ignacio Aedo Cuevas

aedo@ia.uc3m.es

Universidad Carlos III de Madrid
Leganés, Madrid, Spain

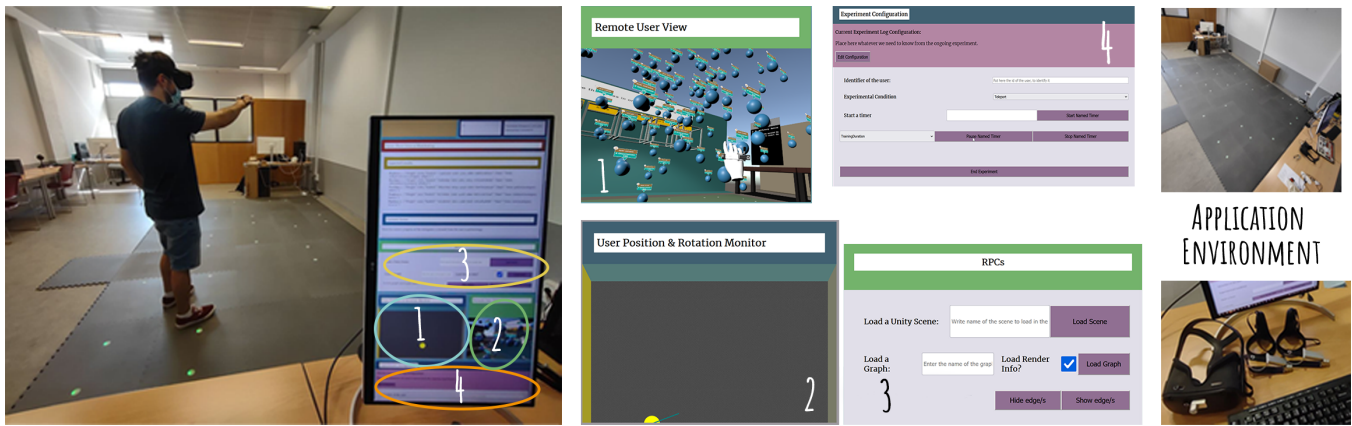


Figure 1: Summary of the developed WOZ tool capabilities (visible in the shown web UI) and sample application environment.

ABSTRACT

We present an open source tool which provides commonly required functionality in HCI experimental research targeted at Immersive technologies. Using the tool, researchers can monitor and control a Unity scene from a controller panel running in the web-browser. Researchers can also use the tool to track the user status (e.g. see what the user sees) and trigger the execution of remote functions (e.g. alter the Unity Scene, the objects appearing, etc.). Moreover, we show a use case to exemplify how the tool is being used to support our research regarding Immersive Multisensory Interaction in a room-sized environment.

CCS CONCEPTS

• **Human-centred computing** → **Systems and tools for interaction design.**



This work is licensed under a Creative Commons Attribution-NonCommercial International 4.0 License.

UIST '21 Adjunct, October 10–14, 2021, Virtual Event, USA

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8655-5/21/10.

<https://doi.org/10.1145/3474349.3480217>

KEYWORDS

Wizard Of Oz, VE Monitoring, Multisensory interaction, HCI Experiment workflow, Virtual Reality

ACM Reference Format:

Alejandro Rey López, Andrea Bellucci, Paloma Díaz Pérez, and Ignacio Aedo Cuevas. 2021. A Tool for Monitoring and Controlling Standalone Immersive HCI Experiments. In *The Adjunct Publication of the 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21 Adjunct)*, October 10–14, 2021, Virtual Event, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3474349.3480217>

1 INTRODUCTION

New and more powerful standalone devices (e.g. Oculus Quest, Microsoft HoloLens) have been developed recently, which provide better support for the study of immersive multisensory interaction in room-sized environments. Unfortunately, the use of these novel devices in experimental research introduces a set of challenges that need to be addressed:

- C1. Wireless/Standalone devices **require an additional software layer to monitor the user performance** during the experiment (e.g. watching the user in the virtual world, recording, monitoring task progress), and makes it more difficult to assist the user during the experiment or react to

unexpected events in-device without interrupting the experiment in progress.

- C2. **In-device logging hinders the data extraction process**, since an additional step to download all log files has to be performed before experimental data can be analysed. Conventional wired HMDs run application in the desktop environment and thus, allow straightforward access to logs.
- C3. **Testing gets more cumbersome and time-consuming**, as a build (e.g. an Android *apk* if working with Oculus Quest¹) needs to be created and executed in the device to refine and fix errors regarding interaction requirements (which are generally driven by users themselves).

Additionally, experiments in HCI have been shown to accommodate to a **common structure** (e.g. [6, 7]), comprised of a *Training Phase*, *Task Execution Phase* and a *Self-reflection/Interview Phase*.

In this context, researchers often make use of a paradigm known as **Wizard of Oz**², which allows studying the user behaviour without requiring a fully-functional implementation of the system or experimental task at hand. Using this approach, researchers can save development time and control the flow of the experiment (e.g. change between phases, switch between scenarios, trigger events, etc.) from a control panel that simulates the functionality of the immersive application and task under study.

To overcome the aforementioned challenges imposed by standalone devices, leverage their advantages in experimental HCI research and provide reusable functionality to support carrying out experiments, we present a client-server application inspired by Wizard of Oz systems which provides a solution for such problems.

2 RELATED WORKS

There have been efforts to create frameworks and middleware to allow the academic community to reuse previous work to handle for them common experimental tasks (e.g. remote control, logging, monitoring, etc.) [1–3, 5]. However, previous attempts are quite domain specific, and focus on desktop interactive applications or non-immersive devices, so they do not adapt painlessly to immersive interactive systems, nor integrate with tools that ease their construction, as it is the case of the Unity³ game engine.

3 SYSTEM OVERVIEW

The proposed tool has been designed to provide a solution to the aforementioned challenges (C1,C2,C3) and with the following premises in mind:

- **Making commonly used functionality readily available for HCI researchers** using Unity to design their interactive scenarios. User tracking, remote procedure execution and log configuration are examples of software functionality required by most experimental setups (C1,C2).
- **Fostering exploratory prototyping with users.** Tweaking game parameters during a single session allows trying

different designs, scenarios or interaction modes in less time-consuming user testing sessions (C3).

- **Flexibility.** The application is structured in a way that adding new messages with custom purposes is easy, and the use of *json* as format for the exchanged data makes it less cumbersome to interpret for novices.
- **Portability.** The setup required for experiments using this tool is minimalistic, it only requires computer (running a web client), a server (running the server side of the app) and a set of HMDs to be monitored (with internet connection).

Figure 2 depicts the architecture of the tool, which comprises 3 main components :

- (1) A C# Websocket Client and a set of utilities in the shape of Unity components. These can be added to a Unity GameObject to start using them for communication between the Immersive Environment and the outer world.
- (2) A website which acts as monitor/control panel for the researcher. It allows watching the user perspective virtual environment and allows execution of remote procedure calls within the immersive application (e.g. switching between scenes, triggering events, start/stop logging functionality, etc.) .
- (3) An *Node.js*⁴ http server. It acts as glue technology between Immersive standalone devices (running a Unity application) and the experiment control panel (running in a web-browser).

The code for the tool is available as two separate Github repositories (js⁵, C#⁶), which also contain instructions on how to get started.

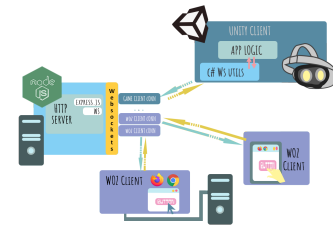


Figure 2: Architecture of the WOZ tool

4 USE CASE/APPLICATION DOMAIN

Our research group focuses on Immersive Analytics, which attempts to unveil the possible benefits of the use of immersive technologies (VR/AR/MR) to perform Data Analysis, extract conclusions and take better informed decisions.

For our experiments, we have an augmented room at our disposal and we make use of standalone HMDs (e.g. Oculus Quest) to provide a cable-free experience to experiment participant. Participants are asked to perform tasks related to IA⁷ in a room-sized VR environment (e.g. interacting with a node-link graph and extract some insights). You can see the usual setup in 1.

The proposed tool has sped up the process of refining and solving

¹<https://www.oculus.com/>

²Wizard Of Oz: technique which allows evaluating incomplete software products by simulating functionality that is not yet available by means of a hidden actor or wizard that bridges user actions and system actions, creating the illusion of a fully-functional system which responds to user input as the final product would [4]

³<https://unity.com/es>

⁴<https://nodejs.org/es/>

⁵<https://github.com/coredamnwork/ws-serverclient-msii>

⁶<https://github.com/coredamnwork/msii>

⁷Immersive Analytics

interaction issues before experiment participant recruitment. Additionally, we leveraged the tool to tweak configuration parameters (e.g. thresholds, animation duration, etc.) during pilot testing. On top of that, during experiment execution, we use the tool's remote control capabilities to switch between experiment phases, assist users (if necessary), and track task progress or configure logging status for a given participant (e.g. start/pause timers, show on demand help, add custom annotations to logs, etc.). Finally, an interesting feature of the tool is that multiple experimenters can watch the experiment and exert control on it. Thus, different researchers in charge of the experiment can focus on specific aspects of the analysis to be performed, even from different devices or locations.

5 CONCLUSION AND FUTURE WORK

We have presented an open source tool to help HCI experimental researchers craft Immersive Interaction experiments and carry them out using standalone Immersive Hardware.

We are currently extending the functionality of the tool to better support Immersive collaboration scenarios, where several users share the same virtual space and control needs to be user-specific. In addition, we will soon simplify the reuse of the tool by distributing code as a Unity package or *npm*⁸ package.

Finally, we plan to simplify the process of adapting the system to researcher needs to make the tool more appealing to non-technical

profiles and analyze the performance of the tool to improve it, if required.

ACKNOWLEDGMENTS

This work is supported by the Spanish State Research Agency (AEI) under grant Sense2MakeSense (PID2019-109388GB-I00).

REFERENCES

- [1] Carmelo Ardito, Paolo Buono, Maria Francesca Costabile, Rosa Lanzilotti, and Antonio Piccinno. 2009. A tool for Wizard of Oz studies of multimodal mobile systems. In *2009 2nd conference on human system interactions*. IEEE, 344–347.
- [2] Jack Brookes, Matthew Warburton, Mshari Alghadier, Mark Mon-Williams, and Faisal Mushtaq. 2020. Studying human behavior with virtual reality: The Unity Experiment Framework. *Behavior research methods* 52, 2 (2020), 455–463.
- [3] Steven Dow, Jaemin Lee, Christopher Oezbek, Blair MacIntyre, Jay David Bolter, and Maribeth Gandy. 2005. *Wizard of Oz Interfaces for Mixed Reality Applications*. Association for Computing Machinery, New York, NY, USA, 1339–1342. <https://doi.org/10.1145/1056808.1056911>
- [4] John F Kelley. 1984. An iterative design methodology for user-friendly natural language office information applications. *ACM Transactions on Information Systems (TOIS)* 2, 1 (1984), 26–41.
- [5] Mirko Otto, Rafael Friesen, and Dietmar Rösner. 2011. Message oriented middleware for flexible wizard of oz experiments in hci. In *International Conference on Human-Computer Interaction*. Springer, 121–130.
- [6] Nico Reski and Aris Alissandrakis. 2020. Open data exploration in virtual reality: a comparative study of input technology. *Virtual Reality* 24, 1 (2020), 1–22.
- [7] Mary K Young, Graham B Gaylor, Scott M Andrus, and Bobby Bodenheimer. 2014. A comparison of two cost-differentiated virtual reality systems for perception and action tasks. In *Proceedings of the ACM Symposium on Applied Perception*. 83–90.

⁸Node Package Manager <https://www.npmjs.com/>